

TRINETRA AI (PRAVAH)

DAM & RESERVOIR INTELLIGENCE MODULE

Master Technical Architecture, Sensor Fusion, OpenCV & Hackathon Blueprint
Focus Sectors: Uttarakhand & Assam

1. Executive Summary & Architecture

The Dam & Reservoir Intelligence Module extends **Trinetra AI** from a general flood-prediction system into an infrastructure-aware disaster warning and decision-support platform. It treats major dams as dynamic hydrological control points, coupling reservoir storage, spillway gate releases, and OpenCV computer vision telemetry directly into the Flash Flood Risk Engine.

CORE CHAIN: Reservoir Telemetry + OpenCV Vision → Upstream Inflow → Gate Outflow Discharge → Downstream Propagation → Context-Aware XGBoost Risk → GIS Safe Evacuation Route

2. Dual-Branch Analytical Architecture

To comply with Central Water Commission (CWC 2018) guidelines, the system separates structural safety from hydrological flood propagation:

- Branch A: Structural Health Branch (Dam Safety Anomaly):** Tracks surface cracks via OpenCV, seepage rate (Lps), and seismic vibration. Generates isolated structural alerts for dam engineers without causing false flood panics.
- Branch B: Hydrological Propagation Branch (Flood Warning):** Ingests live reservoir level (% FRL), inflow, spillway gate openings, and outflow discharge (Cusecs) to compute downstream inundation risk.

3. Target Dam Infrastructure Matrix (Uttarakhand & Assam)

Dam Name	State / River	FRL / Capacity	Downstream Target	Lead-Time Window
Tehri Dam	Uttarakhand (Bhagirathi)	830.0m (FRL)	Rishikesh / Haridwar	4.0 - 5.5 Hours
Tapovan Vishnugad	Uttarakhand (Dhauliganga)	Run-of-River HE	Chamoli / Joshimath	1.0 - 2.5 Hours
Maneri Bhali	Uttarakhand (Bhagirathi)	Run-of-River Stage II	Uttarkashi Town	1.5 - 3.0 Hours
Ranganadi HE Project	Assam / Upstream (NEEPCO)	567.0m (FRL)	Lakhimpur District	2.5 - 4.0 Hours
Subansiri Lower HE	Assam (Subansiri River)	205.0m (FRL)	Dhemaji District	2.0 - 3.5 Hours
Kurichhu HE Plant	Bhutan / Assam Border	530.0m (FRL)	Barpeta / Lower Assam	3.0 - 5.0 Hours

4. OpenCV Computer Vision Integration (`dam\_vision\_detector.py`)

Trinetra AI uses Computer Vision (OpenCV) for non-contact visual dam surface inspection:

- Concrete Surface Crack Detection:** Converts CCTV/Drone video frames to Grayscale, applies Gaussian Blur & Canny Edge Detection, and measures contour bounding geometry to compute crack width (in mm).

2. Visual Gauge Reading: Applies HSV color thresholding to read reservoir water staff gauge height visually as a backup to ultrasonic transducers.

5. Context-Aware XGBoost Risk Coupling

Rather than using a naive rule like 'Gate Open = Flood', the model evaluates the **Combined Context** of dam discharge, downstream river stage, and rainfall intensity. If downstream rivers are already at bankfull stage, an emergency spillway discharge of 45,000 Cusecs rapidly escalates the Flash Flood Risk score from 25% to **92% (CRITICAL ALERT)**.

6. Hackathon 3-Minute Viva Demo Script for Jury

Time Window	Demonstration Action	System Output & Dashboard Reaction
0:00 - 0:45 min	Select Uttarakhand (Chamoli) or Assam (Lakhimpur)	Show baseline state: Dam Level 65%, Gates CLOSED, Risk: LOW (22%).
0:45 - 1:30 min	Drag Inflow Slider to 50,000 Cusecs	Reservoir level rises to 92% FRL. Yellow Watch Alert triggered.
1:30 - 2:30 min	Click 'Open 4 Spillway Gates (45,000 Cusecs Outflow)'	Flood Risk spikes to 92% RED ALERT. Lead-Time Countdown Banner displays 'Surge arrives in 2h 30m'. Map updates safe routes.
2:30 - 3:00 min	Click 'OpenCV Crack Inspection Demo'	Displays live image bounding box highlighting concrete surface crack (>3.2mm) for structural engineers.

7. Key Viva Presentation Guidelines & One-Line Pitch

- **What NOT to Say:** Do NOT claim 'Gate open always means flood' or 'AI automatically controls dam gates'. Position Trinetra AI strictly as a Decision Support System for human operators.
- **Master One-Line Pitch for Professors / Jury:**

"We monitor reservoir and dam structural state, learn how controlled releases propagate downstream, fuse that data with rainfall and river conditions, predict downstream flood risk with lead time, map impact on infrastructure, and generate human-reviewed response recommendations."